

```
1  const e = require("express");
2  const m = require("mysql2");
3  const j = require("jsonwebtoken");
4
5  const app = e();
6  app.use(e.json());
7
8  const K = "mgit_secret_key";
9
10 const db = m
11     .createPool({
12         host: "127.0.0.1",
13         user: "root",
14         password: "password",
15         database: "college_db_5r4",
16         port: 3306,
17     })
18     .promise();
19
20 // middleware
21 const auth = (req, res, next) => {
22     let t = req.headers["authorization"];
23     t = t && t.split(" ")[1];
24     if (!t)
25         return res
26             .status(401)
27             .json({ message: "Access Denied: No Token Provided" });
28
29     j.verify(t, K, (err, u) => {
30         if (err)
31             return res
32                 .status(403)
33                 .json({ message: "Invalid or Expired Token" });
34         req.user = u;
35         next();
36     });
37 };
38
39 // login
40 app.post("/login", (req, res) => {
41     let { username, password } = req.body;
42
43     if (username === "admin" && password === "password123") {
44         let t = j.sign({ username }, K, { expiresIn: "7d" });
45         return res.json({ token: t });
46     }
47
48     res.status(401).json({ message: "Invalid Credentials" });
49 });
50
51 // GET OR READ
52 app.get("/s_details", auth, async (req, res) => {
53     try {
54         let [r] = await db.query("SELECT * FROM s_details");
55         res.json(r);
56     } catch (e) {
57         res.status(500).json({ message: e.message });
58     }
59 });
60
61 // POST
62 app.post("/s_details", auth, async (req, res) => {
63     let { name, course } = req.body;
64
65     try {
66         let [r] = await db.execute(
67             "INSERT INTO s_details (name, course) VALUES (?, ?)",
68             [name, course],
69         );
70
71         res.status(201).json({ id: r.insertId, name, course });
72     } catch (e) {
73         res.status(500).json({ error: e.message });
74     }
75 });
76
77 // PUT
```

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78 app.put("/s_details/:id", auth, (req, res) => {
79     let { id } = req.params;
80     let { name, course } = req.body;
81
82     try {
83         let [r] = await db.execute(
84             "UPDATE s_details SET name=?, course=? WHERE id=?",
85             [name, course, id],
86         );
87
88         if (!r.affectedRows)
89             return res.status(404).json({ message: "Not found" });
90
91         res.json({ message: "Updated successfully", id });
92     } catch (e) {
93         res.status(500).json({ error: e.message });
94     }
95 });
96
97 // DELETE
98 app.delete("/s_details/:id", auth, async (req, res) => {
99     let { id } = req.params;
100
101     try {
102         let [r] = await db.execute("DELETE FROM s_details WHERE id=?", [id]);
103
104         if (!r.affectedRows)
105             return res.status(404).json({ message: "Not found" });
106
107         res.json({ message: `Student ${id} deleted` });
108     } catch (e) {
109         res.status(500).json({ error: e.message });
110     }
111 });
112
113 app.listen(3000, () => console.log("Server running on port 3000"));
114
115 #####App.js
116 import React from "react";
117 import { BrowserRouter as Router, Routes, Route, Link } from "react-router-dom";
118
119 import Home from "../pages/Home";
120 import Register from "../pages/Register";
121 import Login from "../pages/Login";
122 import About from "../pages/About";
123 import Contact from "../pages/Contact";
124
125 function App() {
126     return (
127         <Router>
128             <nav style={{ padding: "10px", background: "black" }}>
129                 <Link to="/" style={linkStyle}>
130                     Home
131                 </Link>
132                 <Link to="/register" style={linkStyle}>
133                     Register
134                 </Link>
135                 <Link to="/login" style={linkStyle}>
136                     Login
137                 </Link>
138                 <Link to="/about" style={linkStyle}>
139                     About
140                 </Link>
141                 <Link to="/contact" style={linkStyle}>
142                     Contact
143                 </Link>
144             </nav>
145
146             <Routes>
147                 <Route path="/" element={<Home />} />
148                 <Route path="/register" element={<Register />} />
149                 <Route path="/login" element={<Login />} />
150                 <Route path="/about" element={<About />} />
151                 <Route path="/contact" element={<Contact />} />
152             </Routes>
153         </Router>
154     );

```

```
155 }
156
157 const linkStyle = {
158   color: "white",
159   margin: "10px",
160   textDecoration: "none",
161 };
162
163 export default App;
164 #Home.js
165 export default function Home() {
166   return (
167     <div>
168       <h1>Home Page</h1>
169       <p>Welcome to Student Management System</p>
170     </div>
171   );
172 }
173
174 #Register.js
175 export default function Register() {
176   return (
177     <div>
178       <h2>Register</h2>
179
180       <input placeholder="Name" />
181       <br />
182       <input placeholder="Email" />
183       <br />
184       <input type="password" placeholder="Password" />
185       <br />
186
187       <button>Register</button>
188     </div>
189   );
190 }
191
192 #Login.js
193 export default function Login() {
194   return (
195     <div>
196       <h2>Login</h2>
197
198       <input placeholder="Email" />
199       <br />
200       <input type="password" placeholder="Password" />
201       <br />
202
203       <button>Login</button>
204     </div>
205   );
206 }
207
208 #About.js
209 export default function About() {
210   return <h2>About Page</h2>;
211 }
212
213 #Contact.js
214 export default function Contact() {
215   return <h2>Contact Page</h2>;
216 }
217
218
219
```